

# Ground Station API

## Method

### Query device

#### Description:

List all devices by session id.

#### Request:

| Parameter  | Description       | Type    |
|------------|-------------------|---------|
| Session_id | Session id number | Integer |

#### Example:

POST /query?session\_id=1

#### Response:

| Parameter | Description   |
|-----------|---------------|
| Name      | Device's name |

#### Example:

```
{
  "devices": [
    {
      "id": "RCKH",
      "name": "CONSOLE40-RCKH"
    }
  ]
}
```

### Add device

#### Description:

Add device to session.

#### Request:

| Parameter | Description | Type |
|-----------|-------------|------|
|-----------|-------------|------|

|            |                   |         |
|------------|-------------------|---------|
| Session_id | Session id number | Integer |
| Device_id  | Device id         | String  |

**Example:**

POST /add\_device?session\_id=1&device\_id=XXXX

**Response:**

No specific response. It returns an empty success response if completed without error

## Register new device

**Description:**

Register new device to database.

**Request:**

| Parameter | Description | Type   |
|-----------|-------------|--------|
| Device_id | Device id   | String |

**Example:**

POST /register\_device?&device\_id=XXXX

**Response:**

No specific response. It returns an empty success response if completed without error

## Request session id

**Description:**

Get a session id.

**Example:**

POST /new\_session

**Response:**

| Parameter  | Description | Type    |
|------------|-------------|---------|
| Session id | Session id  | Integer |

**Example:**

```
{
  "session_id": 2
}
```

## Start session

**Description:**

Start a session for record flight data.

**Request:**

| Parameter  | Description | Type    |
|------------|-------------|---------|
| Session_id | Session id  | Integer |

**Example:**

POST /start\_session?&session\_id=X

**Response:**

No specific response. It returns an empty success response if completed without error

## Stop session

**Description:**

Stop a session.

**Request:**

| Parameter  | Description | Type    |
|------------|-------------|---------|
| Session_id | Session id  | Integer |

**Example:**

POST /stop\_session?&session\_id=X

**Response:**

No specific response. It returns an empty success response if completed without error

## Query flight record

**Description:**

List decive flight record by session id.

**Request:**

| Parameter  | Description       | Type    |
|------------|-------------------|---------|
| Session_id | Session id number | Integer |
| Device_id  | Device id number  | String  |

**Example:**

POST /query\_flight\_data?=session\_id=1&device\_id=RCKH

**Response:****Example:**

```
{
  "flight_record": [
    {
      "aileronAxis": 0.0,
      "airspeedSel": 0.0,
      "alternator": 0,
```

"altitude": 503.0,  
"altitudeSel": 3000.146480102539,  
"aoa": 0.0,  
"autopilot": 0,  
"avionics": 0,  
"battery": 1,  
"beacon": 0,  
"device\_id": "RCKH",  
"elevatorAxis": 0.0,  
"elevatorTrim": -16.070556640625,  
"flaps": 1,  
"flightDirector": 0,  
"flightDirectorBank": 0.0,  
"flightDirectorPitch": 0.0,  
"fuelPump": 1,  
"fuelSelector": 2,  
"fuelWeight": 127.89115646258503,  
"heading": 106.25099387951195,  
"headingSel": 89.000244140625,  
"ias": 0.0,  
"id": 1,  
"landing": 0,  
"landingGear": 1,  
"latitude": 22.33406666665975,  
"longitude": 114.17351944444449,  
"mach": 0.0,  
"magnetos": 4,  
"maxRPM": 0.9797940254211426,  
"mixtureLever": 100.0,  
"navigation": 0,  
"overspeed": 0,  
"parkingBrake": 0,  
"pitch": -2.4742944817990065,  
"pitot": 1,  
"pressure": 1013.1875,  
"propellerLever": 100.0,  
"roll": 0.8955518715083599,  
"rpm": 0.0,

```
    "session_id": 1,  
    "sideSlip": 0.0,  
    "slipSkid": 0.0,  
    "stall": 0,  
    "status": "pausing",  
    "strobes": 0,  
    "taxi": 0,  
    "throttleLever": 58.59375,  
    "time": "Fri, 07 Jan 2022 11:35:44 GMT",  
    "turnRate": 0.0,  
    "verticalSpeed": -417.538153125,  
    "whiskeyCompass": 0.0  
  },  
]  
}
```